

MD. RESHAD AL MUTTAKI

Mirpur 11, Dhaka-1212, Bangladesh
realreshad@gmail.com | +8801919482265 | Portfolio

EDUCATION

Bachelor of Science in Electrical and Electronic Engineering BRAC University Relevant coursework in Electrical and Electronic Engineering	<i>Jan 2022 – Jan 2026</i> Major: Electronics CGPA: 3.81
Higher Secondary Certificate (HSC) Square School & College Rajshahi Board	<i>Passing Year: 2020</i> Major: Science GPA: 5.00
Secondary School Certificate (SSC) Square School & College Rajshahi Board	<i>Passing Year: 2018</i> Major: Science GPA: 5.00

PROFESSIONAL EXPERIENCE

Research Assistant (RA) *Mar 2026 – Present*
BRAC University

- Ongoing research on “Advancing Sustainable Biosensing: A Self-powered Biosensor for Real-time and Wireless Health Monitoring,” funded by RSGI of BRAC University.
- Developing an e-skin-type biosensor both in simulation and hardware environments.

Research Fellow *May 2026 – Present*
BRAC University

- Ongoing research on “GaN-Based G4FET Technology: Fabrication, Characterization, Modeling, and Circuit Demonstration: A Strategic Pathway to Next-Generation Semiconductor Innovation,” funded under the CREST initiative.
- Developing a GaN-Based G4FET up to the fabrication process.

Student Tutor (ST), BSRM SoE, Dept. of EEE *Jan 2024 – Oct 2025*
BRAC University

- Managed and mentored undergraduate students through tutorial and consultation sessions across multiple courses.

Student Mentor, FYAT, OAA *Jan 2024 – Jan 2025*
BRAC University

- Provided one-on-one mentorship to students as part of the First Year Advising Team (FYAT).

COURSES TAUGHT AS STUDENT TUTOR (ST)

Electronic Circuits I (EEE/ECE205) *Spring 2024*
Supervised By: Dr. Mohammed Belal Hossain Bhuiyan

Electrical Circuits I (EEE/ECE101) *Summer 2024*
Supervised By: Atib Mohammad Oni, Mohammad Zunaed

Energy Conversion (EEE221) *Fall 2024 – Summer 2025*
Supervised By: Dr. A.H.M. Abdur Rahim, Mohaimenul Islam, Dr. Shahidul Islam Khan, Shameem Hasan

RESEARCH INTERESTS AND FIELDS

- Artificial Intelligence (AI) & Machine Learning (ML)
- Cybersecurity
- Digital Twin Systems
- Electronics & Intelligent Embedded Systems
- VLSI Design & Hardware Acceleration
- IoT-Based Smart Monitoring Systems
- Cyber-Physical Systems
- Edge AI & Intelligent Sensing
- Semiconductor & Nanoelectronics Research
- Healthcare Technology & Biomedical AI

ACHIEVEMENTS AND HONORS

- **Vice Chancellor’s (VC’s) List** — Spring 2023; Summer 2023; Summer 2024
- **Dean’s List** — Fall 2022; Fall 2023; Spring 2024; Fall 2024; Spring 2025

PUBLICATIONS

Towards smarter grids: A systematic review of wide area monitoring enhancing stability, protection, and promoting environmental sustainability

Energy Conversion and Management: X (Q1, Top 7%) | Impact Factor: 8.78 | CiteScore: 11.3

- DOI: <https://doi.org/10.1016/j.ecmx.2025.101424>

AI-powered cybersecurity for smart grid communication: A systematic review of intrusion detection and threat mitigation systems

Energy Conversion and Management: X (Q1, Top 7%) | Impact Factor: 8.78 | CiteScore: 11.3

- DOI: <https://doi.org/10.1016/j.ecmx.2025.101416>

State-of-charge-aware charging opportunity detection for electric vehicles using data-driven learning and digital twin simulation

Energy Conversion and Management: X (Q1, Top 7%) | Impact Factor: 8.78 | CiteScore: 11.3

- DOI: <https://doi.org/10.1016/j.ecmx.2026.101775>

The next generation of energy storage for smart and sustainable power grids: A systematic review

Energy Conversion and Management: X (Q1, Top 7%) | Impact Factor: 8.78 | CiteScore: 11.3

- DOI: <https://doi.org/10.1016/j.ecmx.2026.101771>

Demand-based anomaly detection algorithm in electric vehicle charging systems using machine learning and digital twin simulation

Journal of Energy Storage (Q1, Top 6%) | Impact Factor: 9.8 | CiteScore: 13.3

- DOI: <https://doi.org/10.1016/j.est.2026.122795>

SKILLS

- | | | | |
|----------------|-----------------|-----------------------|-------------------|
| • Python | • C Programming | • COMSOL Multiphysics | • Git & GitHub |
| • MATLAB | • HTML | • Silvaco TCAD | • Google Suite |
| • Google Colab | • Cadence | • Altium Designer | • MS Office Suite |

REFERENCES

Dr. Mirza Rasheduzzaman

Associate Professor

Department of EEE, BRAC University

mirza.rasheduzzaman@bracu.ac.bd

SK Tahmed Salim Rafid

Lecturer

Department of EEE, BRAC University

rafid.salim@bracu.ac.bd